

Subgroups (Section 5)

$\forall a, b \in H$ we have
 $a * b \in H$

Def: A **subgroup** H of a group G is a subset, closed under the operation $*$ of G , such that $\langle H, * \rangle$ is a group.

We write **$H \leq G$** . (subset $H \subseteq G$)

Ex: $H = \langle \mathbb{Z}, + \rangle$, $G = \langle \mathbb{Q}, + \rangle$ ✓

Non-Ex: $H = \langle \mathbb{Q}^+, \cdot \rangle$, $G = \langle \mathbb{R}, + \rangle$

Not a subgroup, +
must use the induced
operation from G .

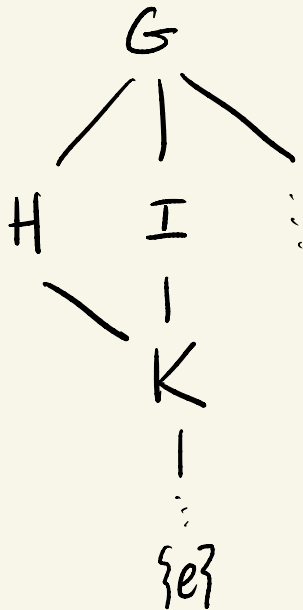
Ex: $\{e\} < G$ **proper subgroup**: $H \leq G$ and $H \neq G$.
trivial subgroup

$$G \leq G$$

Thm: If $K \leq H$ and $H \leq G$, then $K \leq G$.

Pf: Since $K \subseteq H \subseteq G$ we have $K \subseteq G$. Since $K \leq H$ we know K is closed under $*$, and $\langle K, * \rangle$ is a group, so $K \leq G$. $\square$

Subgroup diagrams



↑ bigger
↓ smaller

Group(s) of order 2 & 3

We call the number $|G|$ of elements in a finite group G its **order**.

$$G: \begin{array}{c|cc} & e & a \\ \hline e & e & a \\ \hline a & a & e \end{array}$$

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$$\langle \mathbb{Z}_2, +_2 \rangle \text{ is}$$

a group of order 2

$$\begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0 & 1 \\ \hline 1 & 1 & 0 \end{array}$$

$$\begin{array}{c|ccc} & e & a & b \\ \hline e & e & a & b \\ \hline a & a & b & e \\ \hline b & b & e & a \end{array} : G$$

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$$\langle \mathbb{Z}_3, +_3 \rangle$$

No non-trivial subgroups

	e	a
e	e	a
a	a	e

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

$b \times b = a$
so $\{e, b\}$
not closed

Groups of order 4

$\mathbb{Z}_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\{0, 2\}$ is closed. This is a subgroup isomorphic to $\mathbb{Z}_2$.

	0	1
0	0	1
1	1	0

"Klein 4-group"

V_4	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

$\rightarrow \mathbb{Z}_2$

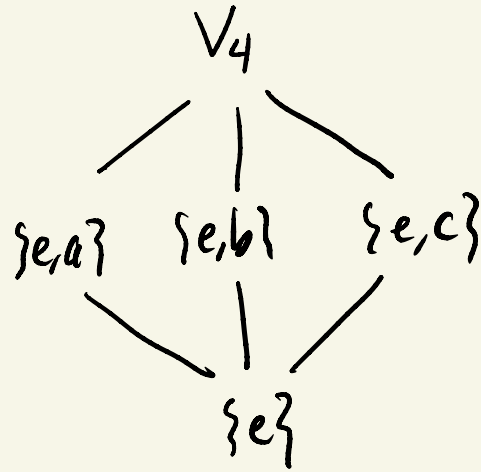
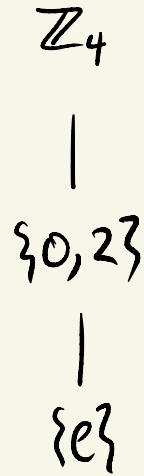
$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

We will see:
If G is finite
& $H \leq G$, then
 $|H|$ divides $|G|$.

What are the subgroups of each?

subgroup isomorphic to $\mathbb{Z}_2$

Subgroup diagrams of $\mathbb{Z}_4$ & V_4



Subgroup diagram is a structural property.

Checking for Subgroups

Thm: A subset H of a group G is a subgroup iff:

- (1) H is closed under the operation of G
- (2) the identity element e of G is in H
- (3) for any $a \in H$, the inverse a^{-1} is also in H .

Pf:

- The identity of $\langle H, * \rangle$ must be e , the identity of G , otherwise $e_H * h = h$, so $e_H * h * h^{-1} = h * h^{-1} = e$
" "
- The inverse of $a \in H$ is a^{-1} e_H
 $a * a^{-1} = e$ then $a^{-1} * a * a^{-1} = a^{-1} * e \rightsquigarrow a^{-1} = a^{-1}$.
- Associativity follows from associativity on G .

The special linear group

Recall: general linear group $GL_n(\mathbb{C}) := \left\{ \begin{array}{l} n \times n \text{ invertible matrices} \\ \text{with } \mathbb{C}\text{-entries} \end{array} \right\}$
under matrix multiplication.

Def: $SL_n(\mathbb{C}) := \left\{ n \times n \text{ matrices with } \mathbb{C}\text{-entries and determinant } 1 \right\}$

Prop: $SL_n(\mathbb{C}) \leq GL_n(\mathbb{C})$.

Pf: $SL_n(\mathbb{C}) \subseteq GL_n(\mathbb{C})$ since $\det \neq 0 \Rightarrow$ invertible.

① If $\det(A) = \det(B) = 1$, then $\det(AB) = \det(A)\det(B) = 1$,
so $SL_n(\mathbb{C})$ is closed.

② $\det(I) = 1$, so $SL_n(\mathbb{C})$ contains the identity

③ $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$, so $SL_n(\mathbb{C})$ is closed under
inverses.

Cyclic subgroups

For $a \in G$, written multiplicatively, let $a^n = \overbrace{a * \dots * a}^{n \text{ times}}$, $a^{-1} = a'$, and $a^{-n} = (a^{-1})^n$. Let $a^0 = e$.

Thm: Let a be an element of a group G . Then

$$H = \{ a^n \mid n \in \mathbb{Z} \} \leftarrow \text{the cyclic subgroup of } G \text{ generated by } a, \text{ denoted } \langle a \rangle.$$

is a subgroup of G , and every subgroup of G which contains a must contain H .

Pf: ① $a^n * a^m = a^{n+m} \in H$, so H is closed

$$\text{② } e = a^0 \in H$$

$$\text{③ } (a^n)' = a^{-n} \in H \text{ so } H \text{ has inverses.}$$

Def: An element $a \in G$ **generates** G if $\langle a \rangle = G$.

G is **cyclic** if there is some element a that generates it.

$$\{0, 1, 1+1, 1+1+1\}$$

Ex:

$\mathbb{Z}_4$: is cyclic since $\langle 1 \rangle = \mathbb{Z}_4$ (and $\langle 3 \rangle = \mathbb{Z}_4$)

$\mathbb{Z}_n$: is cyclic since $\langle 1 \rangle = \mathbb{Z}_n$

V_4 : is not cyclic.